AI PRODUCT CASE STUDY · COGNIZANT CANADA (CONCEPT)

An onboarding agent that gets consultants contributing in days, not weeks

15 → 12 days
target time to first contribution

31 / 31
broken plans caught by the guard

4.8×
year-1 ROI, base case

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How I worked: six steps over two weeks

1
Discover
Personas, journey, hypotheses

2
Define
PRD, activation metric

3
Experiment
Hypothesis, sample size, stop rules

4
Design
Control vs treatment in Figma

5
Build
Multi-agent prototype on Google ADK

6
Evaluate
10 joiners, 31 seeded errors

Experimentation is the core skill here: the ship decision rests on an A/B test, not on model accuracy.

Onboarding has five owners and no single plan

Joiners

They don’t know the next step. A 7-day client background check often starts on day 4.

Project managers

An estimated 6 hours per joiner answering repeat questions, with no view of who is ready.

The business

Every idle day is a consultant on the bench. At 40 joiners a month, 3 days saved is about 1,400 working days a year.

Baselines are hypotheses from delivery experience, to be replaced by a five-person interview plan.

Three people, one goal: ready to contribute

Aisha, new hire

Toronto · first project in week 2

Wants to be useful fast. Loses time waiting on access and working out the order of steps.

Daniel, project switcher

Montréal · rolling onto a bank client

Repeats setup on every project. Needs client clearances and French documents quickly.

Mei, project manager

Runs a 12-person delivery team

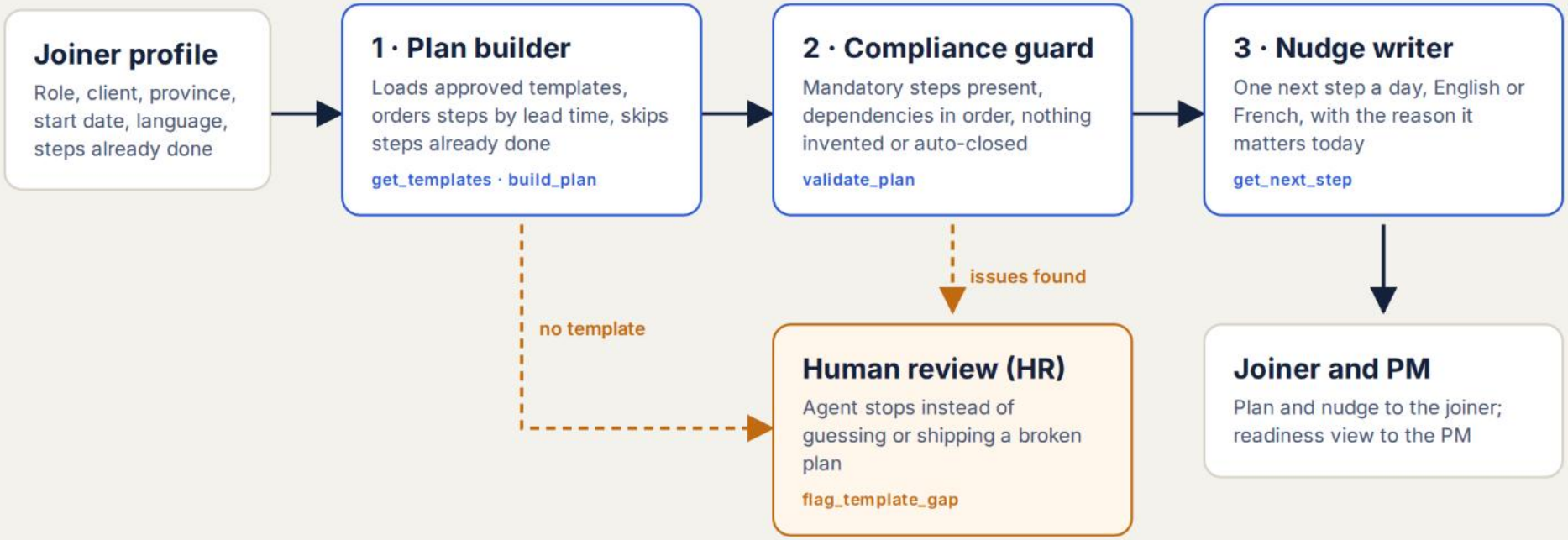
Wants joiners billable in week 1. Has no view of who is blocked and why.

Four hypotheses to test before sizing the experiment

Hypothesis	How it gets checked
Over half the delay is waiting on access and clearances that could start earlier	Interviews and past onboarding records
Joiners lose time not knowing the next step, not because steps are hard	Interviews: “How did you know what to do next?”
PMs spend about 6 hours per joiner on repeat questions	PM time survey
Switchers repeat setup a reusable profile would remove	Switcher interviews

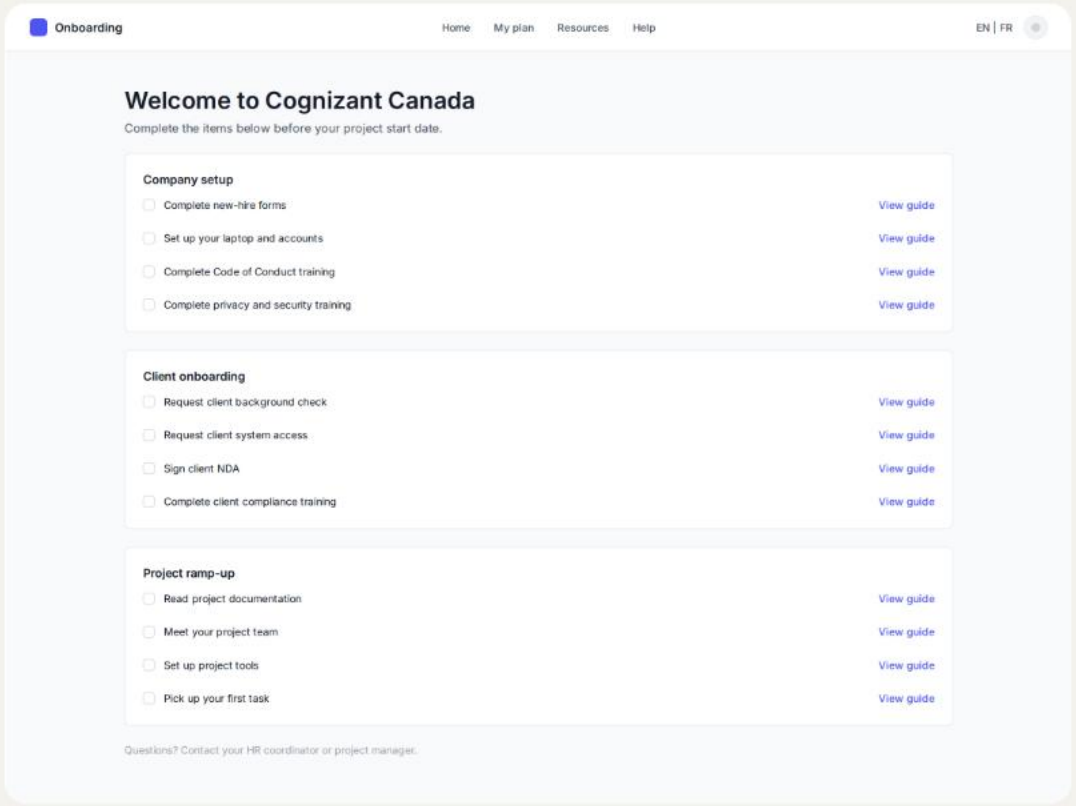
Three agents, two places where a human decides

GOOGLE ADK · GEMINI · DETERMINISTIC RULES IN CODE

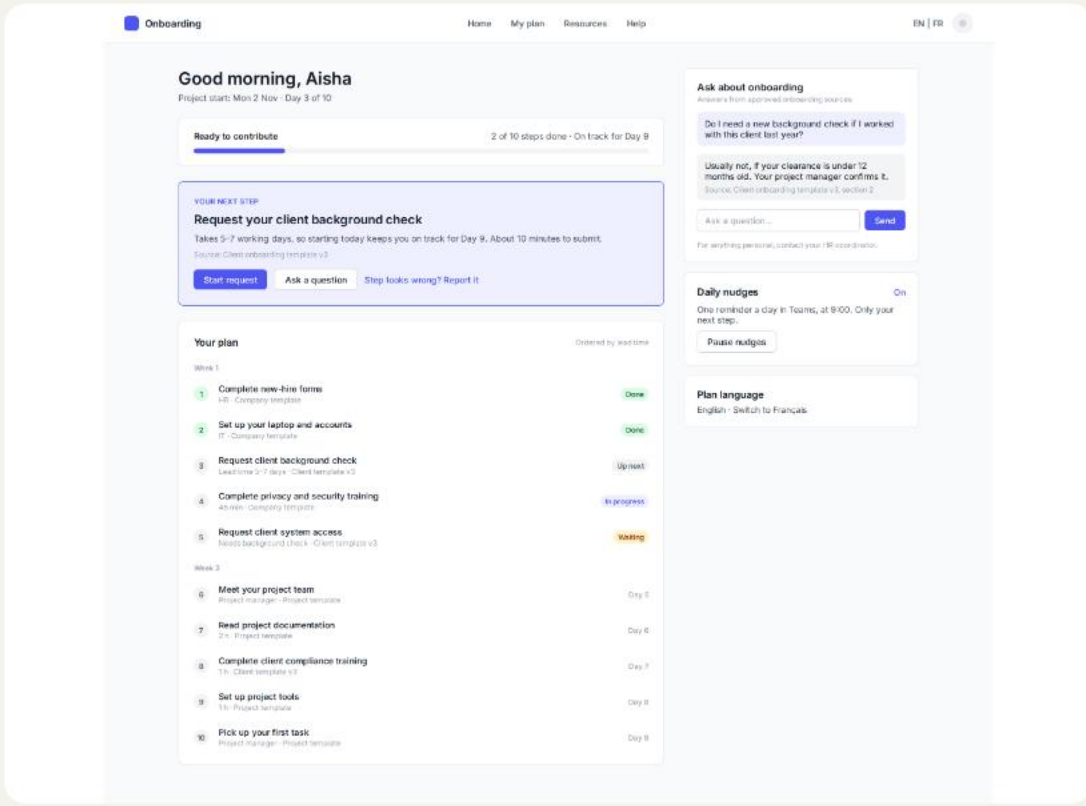


Blue: AI agents · Orange dashed: paths that stop for a human

Control vs treatment: what the A/B test compares

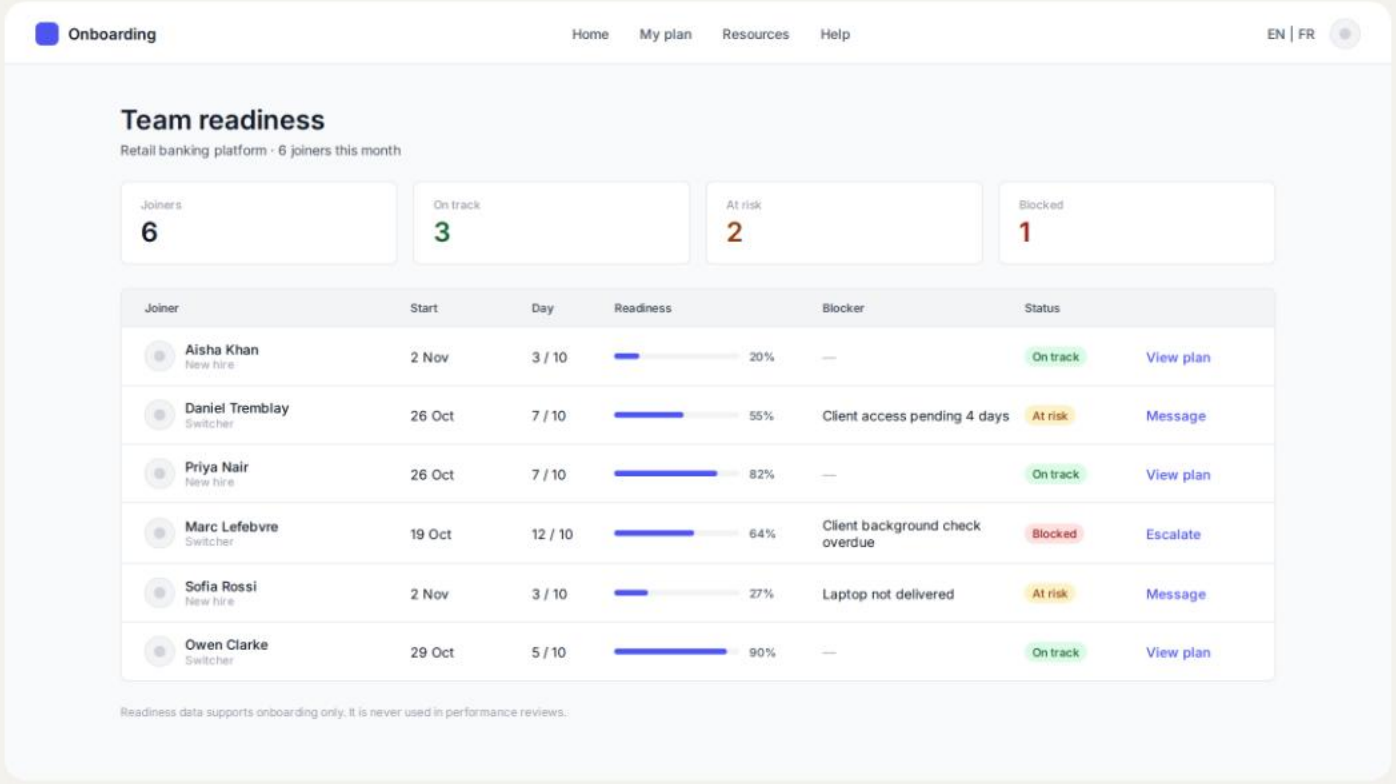


Control: 12 items, no order, no owners

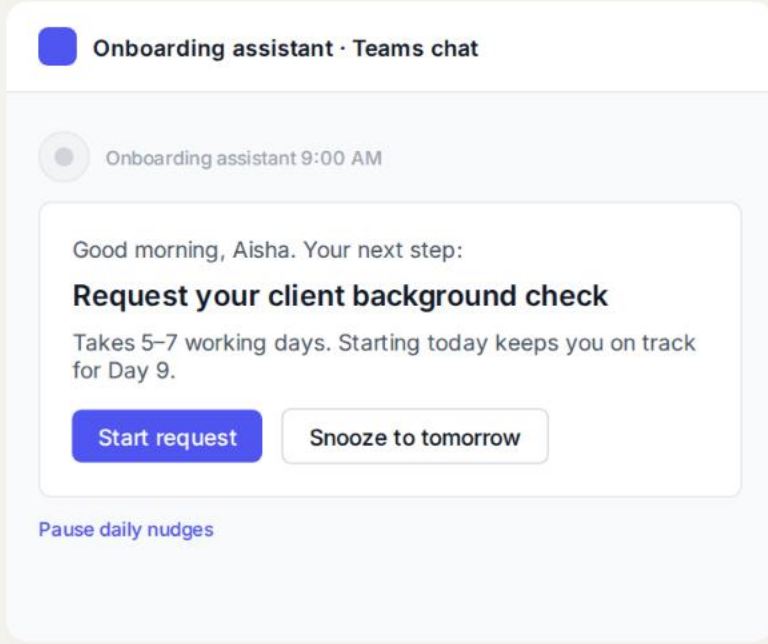


Treatment: one next step with the reason, plan ordered by lead time

Treatment also reaches the PM and the joiner's chat

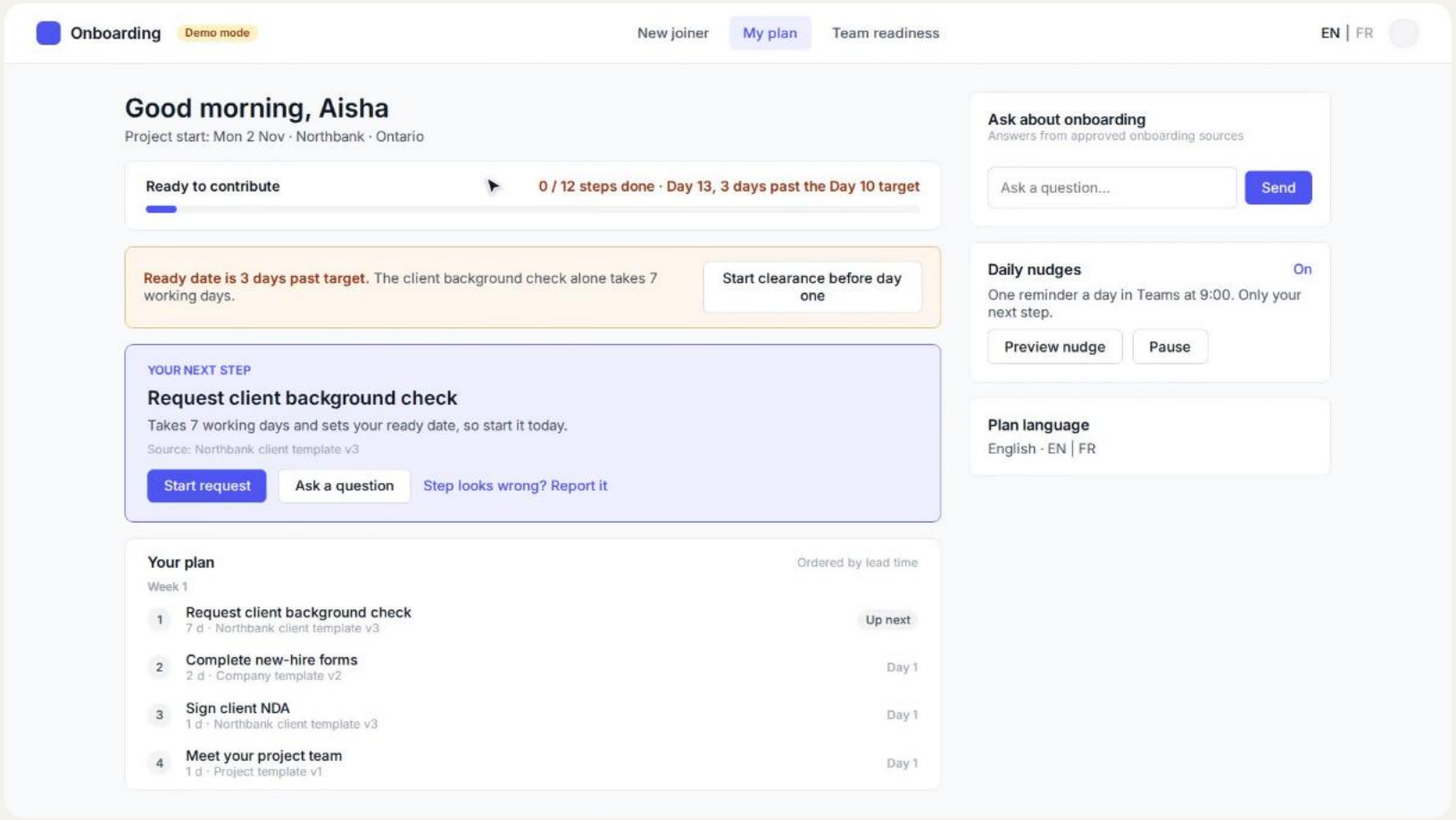


PM readiness view: blockers and one action per joiner



Daily Teams nudge: pausing is one tap

Product demo: one session, end to end



0:00 HR sets up a new hire

0:18 Plan shows Day 13

0:25 Clearance before day one: Day 6

0:40 Q&A with a source

0:52 Teams nudge, then French

1:28 No template: paused for HR

1:40 PM readiness view

Experiment design: powered to decide in 3 months

Hypothesis: a plan ordered by lead time plus a daily nudge cuts time to first contribution from 15 to 12 working days.

Unit	Individual joiner, 50/50, stratified by segment and province
Decision metric	Working days to first contribution
Guardrails	Compliance steps skipped · nudge pause rate · plan errors
Sample	63 per arm, 126 total · about 3 months
Ship rule	≥ 2 days faster, significant, no guardrail breached

The call that mattered

Activation rate (35% → 50%) needs 340 joiners, about 8.5 months. A continuous metric answers in 3. Activation stays as a reported metric.

Assumes 40 joiners a month and a 6-day standard deviation, both to validate before launch.

Agents decide, code enforces the rules

Why agentic

A fixed workflow can’t handle a client with no template. The plan builder chooses: build the plan, or stop and flag HR.

Why rules in code

Mandatory steps, dependency order and “never close a compliance step” are plain Python the agents call as tools.

Stack

Google ADK and Gemini: three agents, five tools. Synthetic templates for two clients plus a Quebec addendum. Streamlit demo with an agent trace.

Evaluation: 10 synthetic joiners, 31 seeded errors



What the eval found

Northbank new hires can’t be ready before **Day 13**: the background check alone takes 7 days. The fix is a process change, start clearance before day one, not a better model.

Seeded errors: dropped mandatory step, steps out of order, invented step, compliance step marked done by the agent.

Business case: 1.0× at worst, 4.8× base

	Low	Base	High
Days saved per joiner	1	3	5
PM hours saved per joiner	1	3	4
Adoption	50%	80%	100%
Year-1 value	$179K	$861K	$1.74M
Return on $180K	**1.0×**	**4.8×**	**9.7×**

Assumptions: 480 joiners a year, 7.5-hour day, $85/h consultant and $110/h manager cost, $180K year-1 build and run cost.

What I learned

Pick a metric you can afford

The best metric is useless if the test takes most of a year. Powering on a continuous metric cut it to 3 months.

Agents need a place to stop

The most valuable behaviour is refusing to guess. The template-gap path makes the agent safe to trust.

Evals find product problems

The Day 13 finding came from the eval, not research. A model question became a process change.

Next: run the 5 interviews, validate the 40-a-month and 6-day assumptions, then pilot with one delivery unit.